EECE.5494 MODEL-BASED SYSTEMS ENGINEERING

OOSEM, MagicGrid

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			Pillar			
Domain			Requirements	Structure	Behavior	Parameters
	Problem	Black Box	Stakeholder Needs	System Context	Use Cases	Measures of Effectiveness
		White Box		Conceptual Subsystems	Functional Analysis	MoEs for Subsystems
	Solution		System Requirements	System Structure	System Behavior	System Parameters
			Subsystem Requirements	Subsystem Structure	Subsystem Behavior	Subsystem Parameters
			Component Requirements	Component Structure	Component Behavior	Component Parameters
	Implementation		Implementation Requirements			

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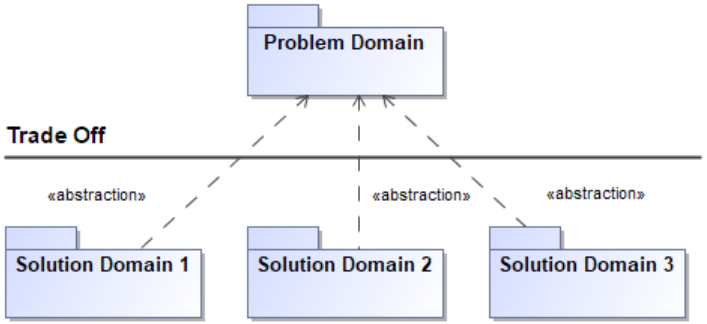
MAGICGRID - SOLUTION

- Develops an accurate model of the system's logical architecture and high-level design.
- Presents one or more solutions for defining the problem domain.
- Enables trade-off analysis to determine the optimal solution for system implementation.



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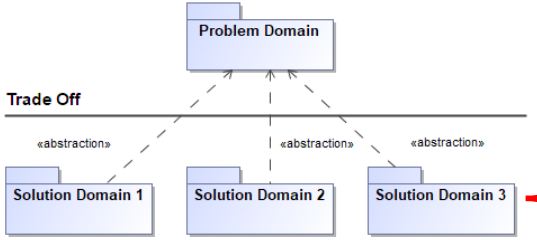
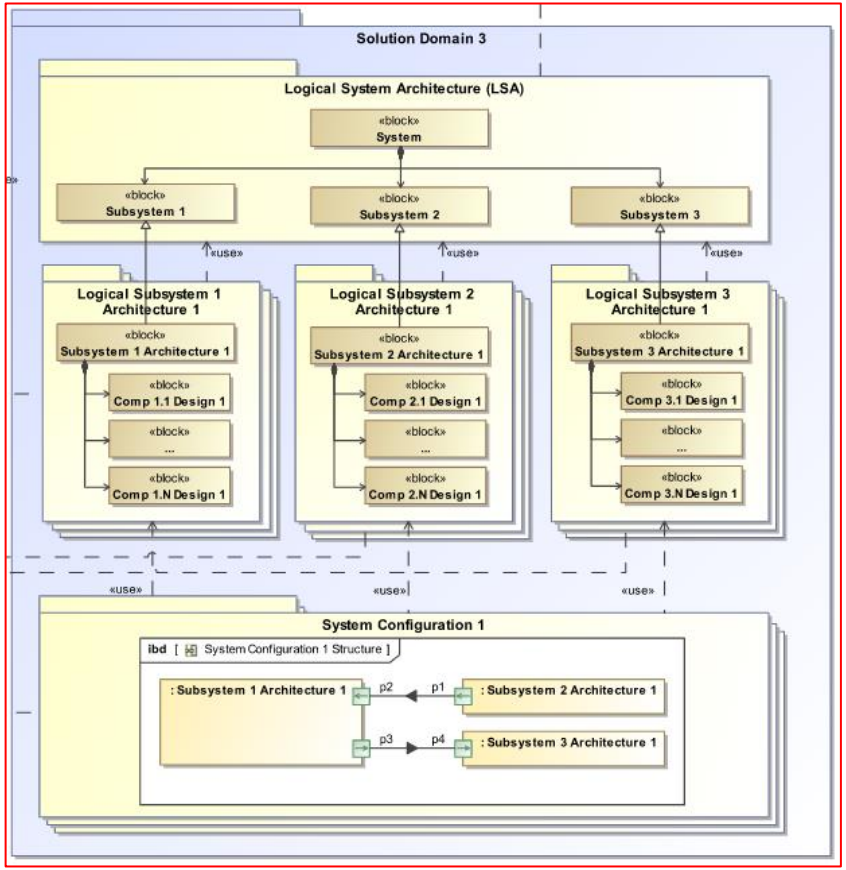
ORGANIZING SOLUTION DOMAIN





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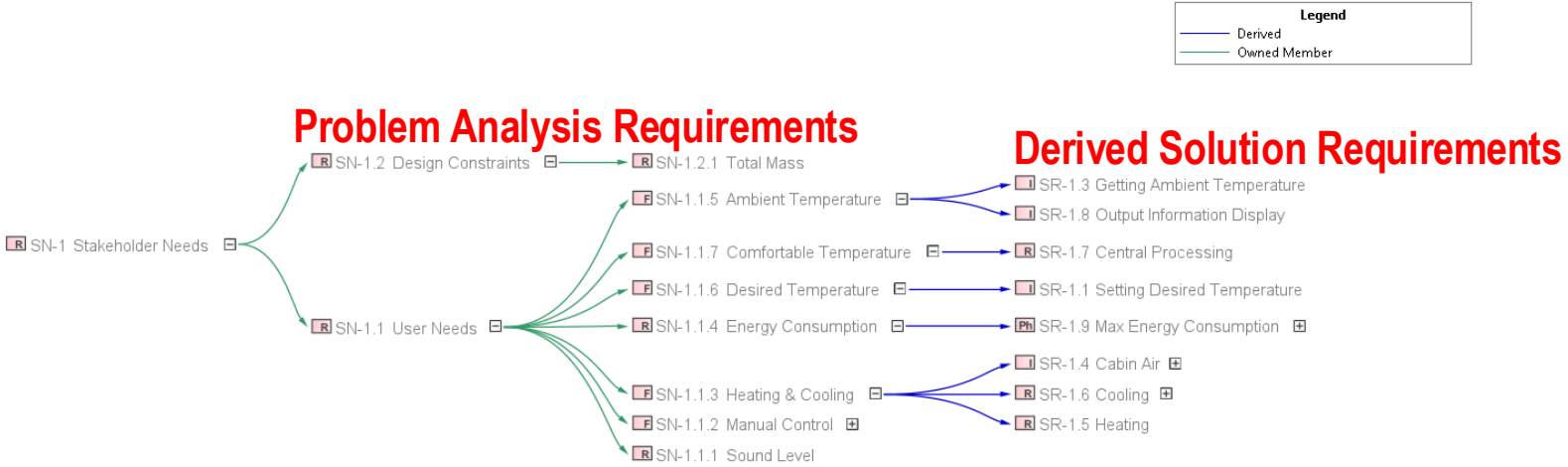
SOLUTION – SYSTEM REQUIREMENTS

- The problem domain analysis yields a SysML model that represents the system requirements specification. [Problem domain analysis derives solution].
- This model can be used as a basis for generating a textual system requirements specification.
- System requirements, similar to stakeholder needs, are uniquely identified.
- These requirements can be organized by features and classified into categories.
- System requirements further define the problem domain and are directly derived from stakeholder needs.



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SOLUTION – SYSTEM REQUIREMENTS TRACEABILITY

Legend

↗ Satisfy

VCCS

		: Energy Consumption	: Max Energy Consumption	/totalEnergyConsVCCS : kWh	cooling : Cooling System	cp : CP System	heating : Heating System	hmi : HMI System	in p1 : UserDataInput	in p2 : SensorDataInput	in p3 : InsideAirInlet	out p4 : InsideAirOutlet	out p5 : OutputInfo	out p6 : OutsideAirOutlet	out p7 : Waste
1 System Requirements		1		1	1	1	1	1	1	1	1	1	1	1	1
SR-1 SRS for VCCS															
SR-1.1 Setting Desired Temperature	1								↙						
SR-1.2 Human-Machine Interactions	1							↙							
SR-1.3 Getting Ambient Temperature	1									↙					
SR-1.4 Cabin Air	2										↙	↙			
SR-1.5 Heating	1						↙								
SR-1.6 Cooling	1				↙										
SR-1.7 Central Processing	1					↙									
SR-1.8 Output Information Display	1												↙		
SR-1.9 Max Energy Consumption	2	↙		↙											
SR-1.10 Hot Air Removal form VCCS	1													↙	
SR-1.11 Waste Removal VCCS	1														↙

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SOLUTION- SYSTEM STRUCTURE TRACED TO BLACKBOX STRUCTURE

Legend

↗ Abstraction

White box system structure analysis

Solution system structure analysis

	Cooling	Heating	Human-Machine Interacting	Processing	VCCU
Cooling System	↗				
CP System				↗	
Heating System		↗			
HMI System			↗		
VCCS					↗

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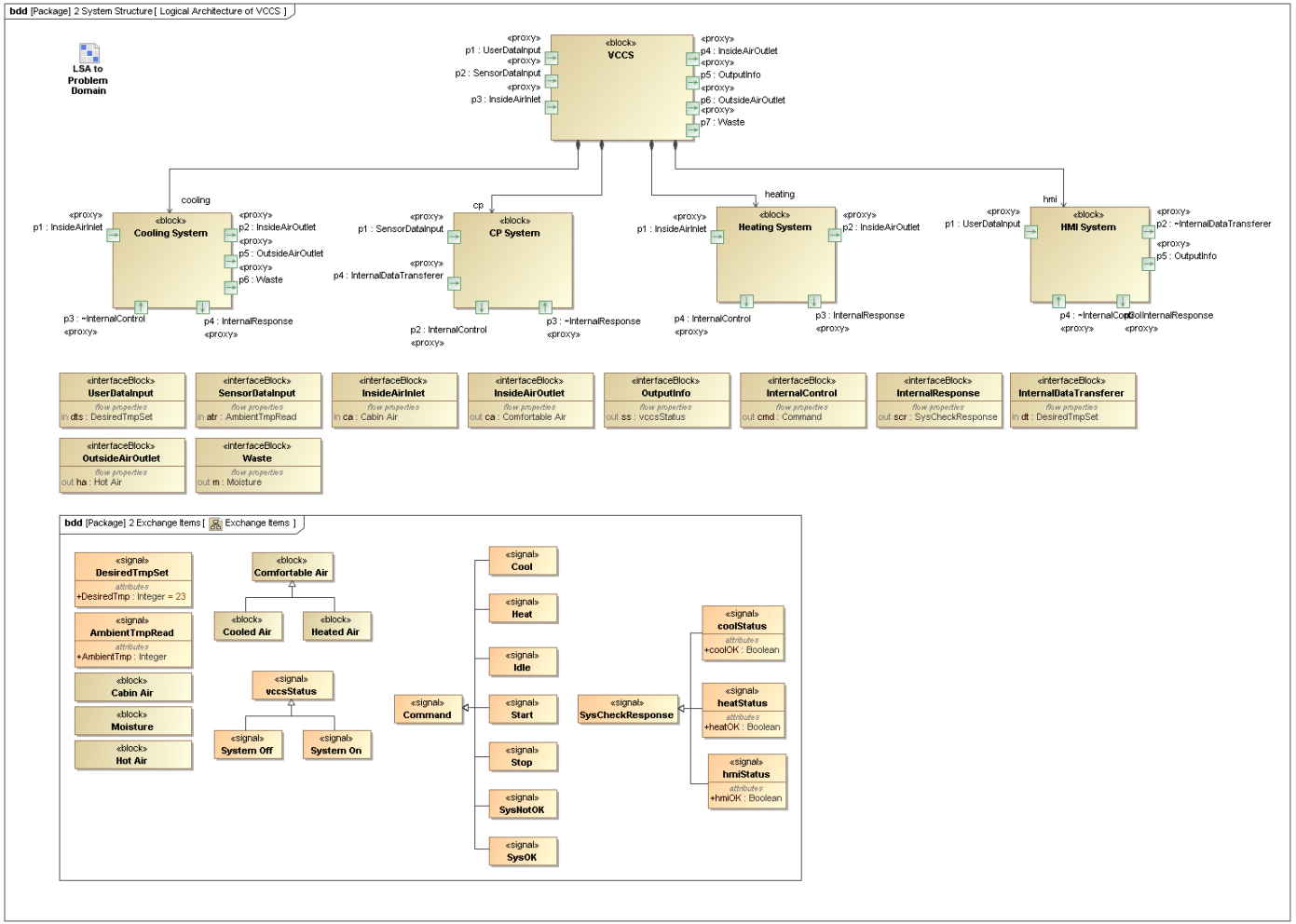
SOLUTION – SYSTEM STRUCTURE



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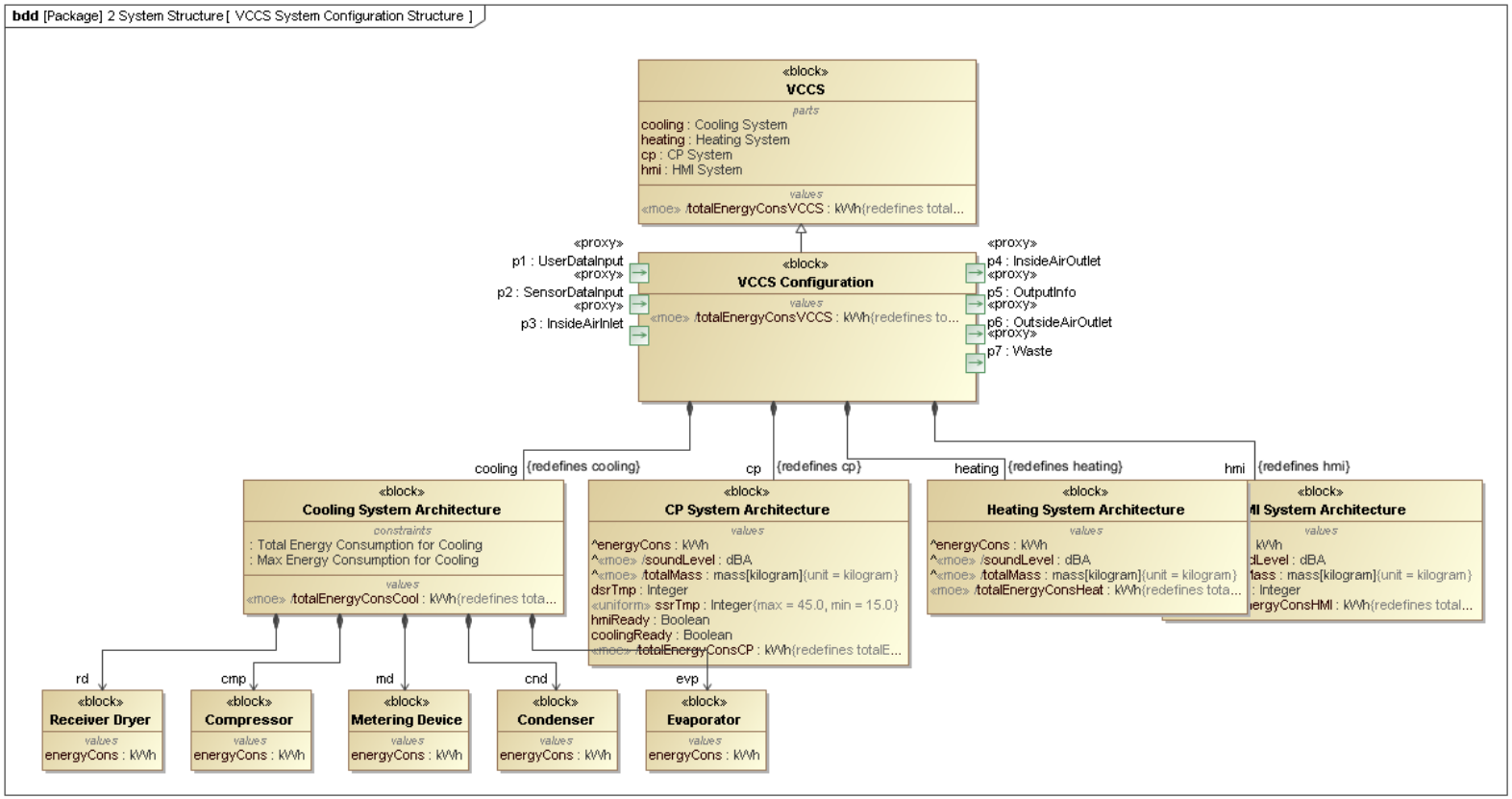
SOLUTION – SYSTEM STRUCTURE



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SOLUTION – SYSTEM BEHAVIOR

Within the context of Logical Structural Analysis (LSA), the focus is on defining the structure and relationships of the system's components rather than on detailing how the system behaves or operates. In other words, LSA is concerned with the architecture and organization of the system, not with specifying the dynamic aspects or interactions that constitute its behavior.

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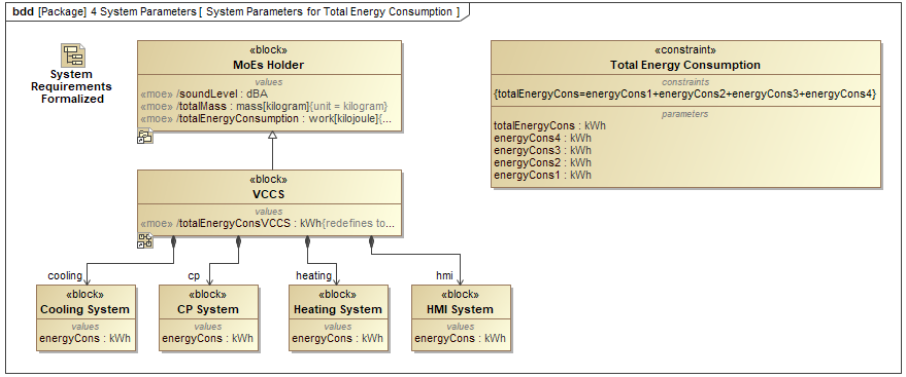
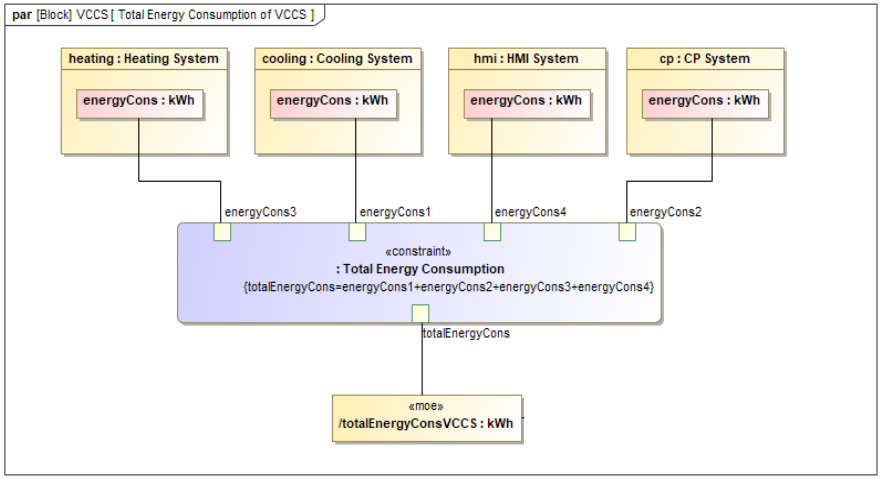
SOLUTION – SYSTEM PARAMETERS

- Quantitative attributes of the System of Interest (SOI) are documented as system parameters.
- These parameters are essential for determining Measures of Effectiveness (MoEs) identified in the problem domain analysis.
- The model can include mathematical rules that outline how these parameters are derived.





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TRACEABILITY FROM LSA TO SYSTEM REQUIREMENTS

Legend

Satisfy

VCCS

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APPLY MAGICGRID RECURSIVELY

MagicGrid at System Level

			Pillar				
			Requirements	Structure	Behavior	Parameters	Safety & Reliability
Domain	Problem	Black Box	Stakeholder Needs	System Context	Use Cases	Measures of Effectiveness (MoEs)	Conceptual and Functional Failure Mode & Effects Analysis (FMEA)
		White Box		Conceptual Subsystems	Functional Analysis	MoEs for Subsystems	Conceptual Subsystems FMEA
	Solution		System Requirements	System Structure	System Behavior	System Parameters	System Safety & Reliability (S&R)
			Subsystem Requirements	Subsystem Structure	Subsystem Behavior	Subsystem Parameters	Subsystem S&R
			Component Requirements	Component Structure	Component Behavior	Component Parameters	Component S&R
	Implementation		Implementation Requirements				

MagicGrid **at Subsystem/Component Level**

				Pillar					
				Pillar					
				Pillar					
				Requirements	Structure	Behavior	Parameters	Safety & Reliability
Domain	Domain	Domain	Problem	Black Box	Stakeholder Needs	System Context	Use Cases	Measures of Effectiveness (MoEs)	Conceptual and Functional Failure Mode & Effects Analysis (FMEA)
				White Box		Conceptual Subsystems	Functional Analysis	MoEs for Subsystems	Conceptual Subsystems FMEA
			Solution		System Requirements	System Structure	System Behavior	System Parameters	System Safety & Reliability (S&R)
					Subsystem Requirements	Subsystem Structure	Subsystem Behavior	Subsystem Parameters	Subsystem S&R
					Component Requirements	Component Structure	Component Behavior	Component Parameters	Component S&R
			Implementation		Implementation Requirements				

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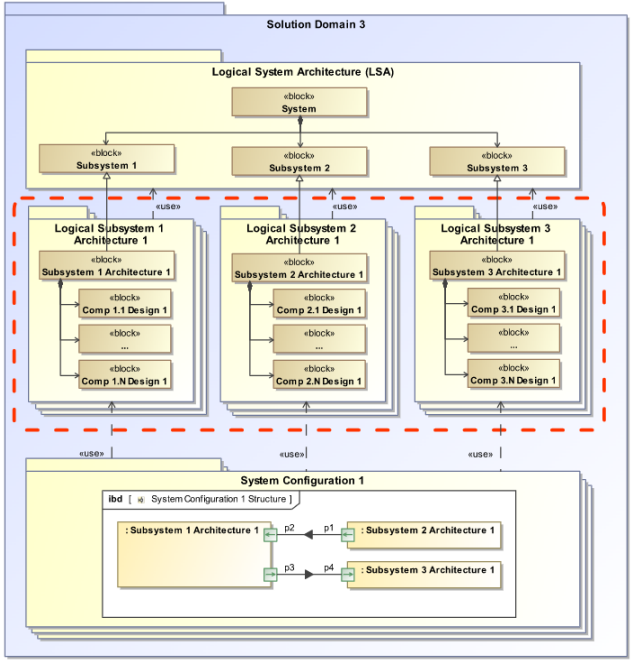
SOLUTION – SUBSYSTEM STRUCTURE

- Different engineering teams concurrently develop logical architectures for various subsystems.
- The process begins by detailing the internal structure of each subsystem and outlining communication between its components.
- Multiple versions of the logical architecture for each subsystem can be created
- One subsystem as an example is cooling system



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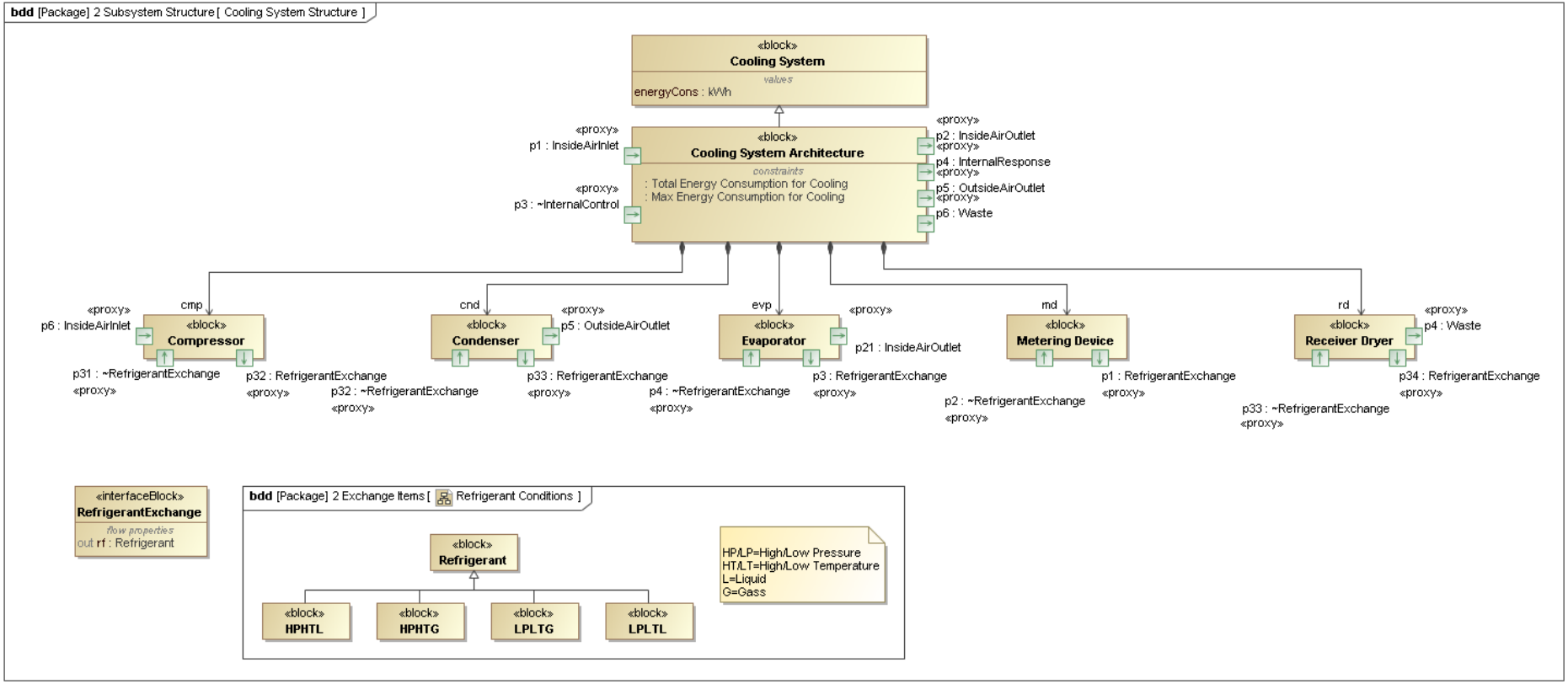
SOLUTION – SUBSYSTEM STRUCTURE – COOLING SYSTEM



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